

MACHINE LEARNING-BASED EQUATION DISCOVERY IN LAND-ATMOSPHERE MODELLING

A Research Project Proposal

Prepared in fulfilment of the academic requirements for the proposed research project

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Project Area: Climate, Land Surface Modelling and Machine Learning

2026

ABSTRACT

Machine learning has increasingly demonstrated the ability to emulate complex environmental processes, yet many high-performing models remain difficult to interpret and therefore offer limited scientific insight. Land–atmosphere modelling presents a particularly important case because exchanges of carbon, water and energy between terrestrial ecosystems and the atmosphere are governed by interacting and nonlinear processes that must be represented in land surface models. This study proposes a machine learning-based equation discovery framework for deriving parsimonious and interpretable relationships for selected land–atmosphere fluxes from meteorological and flux observations. The study will focus on the Sparse Identification of Nonlinear Dynamics (SINDy) framework, which performs sparse regression over a library of candidate terms to identify a compact mathematical representation of observed dynamics. Using FLUXNET observations, physically informed candidate functions will be constructed from meteorological forcings and relevant state variables. Alternative SINDy configurations will then be evaluated against conventional statistical and machine-learning baselines. The research will assess predictive performance, sparsity, physical plausibility, robustness across sites and biomes, sensitivity to noise and missing data, and computational scalability. Particular attention will be given to the extent to which discovered equations generalise beyond the conditions under which they are trained. The expected outcome is a systematic assessment of the capabilities and limitations of equation discovery for land surface modelling and a reproducible computational workflow implemented in Python and supported by high-performance computing and version-controlled software development. The study is expected to contribute evidence on whether

sparse, physically inspired machine learning can produce useful and scientifically interpretable approximations of land–atmosphere exchange processes.

Keywords: equation discovery; SINDy; land–atmosphere exchange; FLUXNET; interpretable machine learning; land surface modelling; sparse regression; eddy covariance.

CHAPTER ONE

INTRODUCTION

1.1 Background to the Study

Land surfaces continuously exchange energy, water vapour and carbon dioxide with the atmosphere. These exchanges influence weather, climate, ecosystem productivity, drought response and the broader terrestrial carbon cycle. Representing them accurately is therefore a central task in land surface modelling. However, the underlying processes are nonlinear and depend on interacting controls such as incoming radiation, air temperature, atmospheric humidity, wind, precipitation, soil moisture and vegetation state. A major scientific challenge is to represent these relationships in a way that is both sufficiently accurate and physically interpretable.

Process-based land models are valuable because their structure is linked to physical and ecological understanding, but they may contain uncertain parameters, simplified process representations and computationally expensive parameterisations. Conversely, machine-learning models can learn flexible relationships from large observational datasets, yet many operate as black boxes. High predictive accuracy alone is not sufficient when the scientific objective includes understanding the mechanisms and functional relationships that generate observed behaviour. Recent work in geosciences has consequently emphasised approaches

that combine data-driven learning with physical knowledge in order to improve interpretability and generalisation (Shen, 2018; Shen et al., 2023).

Equation discovery provides a possible middle ground between purely process-based and purely black-box modelling. Instead of only predicting a target variable, equation discovery seeks a compact mathematical expression that describes the observed relationship. Sparse Identification of Nonlinear Dynamics (SINDy) is one of the most influential methods in this area. Introduced by Brunton, Proctor and Kutz (2016), SINDy constructs a library of candidate nonlinear functions and uses sparse regression to identify a small subset that best explains the data. The resulting model is explicitly symbolic and can therefore be inspected, tested and compared with existing scientific expectations.

The availability of global eddy-covariance observations makes land–atmosphere exchange an appropriate domain for testing such methods. The FLUXNET2015 dataset provides harmonised observations and associated meteorological variables from a large number of sites, together with standardised quality-control and processing procedures (Pastorello et al., 2020). These data offer the opportunity to examine whether physically inspired candidate terms can be combined through sparse equation discovery to recover interpretable relationships for land–atmosphere fluxes across contrasting environmental conditions.

Although equation discovery has expanded rapidly in physics and engineering, its use in land surface modelling remains comparatively limited. This creates both an opportunity and a methodological challenge. Land–atmosphere data are noisy, heterogeneous, non-stationary and influenced by processes operating across multiple temporal scales. Standard SINDy can also be sensitive to measurement noise, derivative estimation and the choice of

candidate library (Kaiser et al., 2018; Messenger et al., 2022). Consequently, the usefulness of equation discovery in this field must be evaluated critically rather than assumed. The proposed study addresses this gap by systematically testing the capabilities and limitations of SINDy-based equation discovery for land–atmosphere flux modelling.

1.2 Statement of the Problem

Machine-learning approaches are increasingly used in environmental modelling because of their ability to capture nonlinear relationships from large datasets. However, many high-performing methods provide predictions without yielding concise mathematical relationships that scientists can readily inspect or incorporate into process-based understanding. In land surface modelling, this lack of interpretability can limit confidence in model behaviour, particularly when predictions are applied under climate conditions that differ from those represented in the training data.

At the same time, existing process-based formulations may be difficult to constrain directly from observations because multiple processes interact and the most appropriate functional form is not always known. Equation discovery could potentially identify compact relationships directly from observations while retaining scientific interpretability. Yet it remains unclear whether SINDy can reliably recover stable and physically meaningful equations from heterogeneous FLUXNET observations. Its sensitivity to noise, candidate-library specification, regularisation choices, temporal resolution and site-to-site variability may substantially affect the equations obtained.

The core problem of this study is therefore the absence of a sufficiently tested and interpretable framework for applying machine learning-based equation discovery to observational land–atmosphere flux data. The study seeks to determine whether SINDy can

identify parsimonious, physically plausible and generalisable functional equations, and under what conditions the method fails or produces unstable results.

1.3 Aim of the Study

The aim of this study is to investigate the capabilities and limitations of machine learning-based equation discovery, with particular emphasis on the Sparse Identification of Nonlinear Dynamics (SINDy) approach, for deriving interpretable functional relationships governing selected land–atmosphere fluxes from FLUXNET observations.

1.4 Objectives of the Study

To review the theoretical foundations and recent applications of machine learning-based equation discovery and SINDy.

To prepare and harmonise selected meteorological, environmental and land–atmosphere flux variables from FLUXNET observations.

To construct a physically informed library of candidate nonlinear terms based on established land–atmosphere controls.

To apply sparse regression and alternative SINDy configurations to discover parsimonious equations for selected land–atmosphere fluxes.

To compare the predictive performance and interpretability of discovered equations with conventional statistical and machine-learning baselines.

To evaluate the robustness of the discovered equations across sites, climatic conditions, vegetation types and data-quality conditions.

To identify methodological limitations associated with noise, missing data, derivative estimation, feature-library design, regularisation and extrapolation.

To develop a reproducible Python-based workflow suitable for high-performance computing and collaborative version-controlled research.

1.5 Research Questions

Can SINDy identify compact and interpretable equations for selected land–atmosphere fluxes from FLUXNET observations?

Which physically inspired meteorological and environmental terms are consistently selected across different sites and biomes?

How does the predictive performance of SINDy-discovered equations compare with conventional regression and machine-learning models?

How sensitive are the discovered equations to data noise, missing observations, candidate-library specification and sparsity thresholds?

To what extent do equations discovered at one set of sites generalise to unseen sites and environmental conditions?

What are the principal computational and scientific limitations of applying equation discovery to land surface modelling?

1.6 Research Hypotheses

H1: A physically informed SINDy framework can identify sparse equations that explain a substantial proportion of variation in selected land–atmosphere fluxes. H2: Equations constrained by physically motivated candidate terms will be more interpretable and more stable than equations derived from unrestricted feature libraries. H3: Predictive performance will vary across flux type, climate and biome, and equation generalisation will decline when models are applied outside the environmental conditions represented in the

training data. H4: Noise, derivative estimation and feature-library specification will be major sources of uncertainty in standard SINDy applications.

1.7 Significance of the Study

Scientifically, the study will contribute to the growing effort to make machine learning more interpretable and physically useful in Earth system science. Rather than treating machine learning only as a predictive instrument, the proposed work evaluates its potential as a tool for scientific discovery. This is particularly relevant where the objective is to identify compact process relationships that can be inspected by domain experts.

Methodologically, the study will provide a systematic comparison of alternative equation-discovery configurations under realistic environmental data conditions. The findings may clarify when standard SINDy is adequate and when more robust approaches, such as ensemble or weak-form formulations, are required. Ensemble-SINDy has been proposed specifically to improve robustness under limited and noisy data conditions, while weak formulations can reduce the direct dependence on derivative estimation (Messenger et al., 2022).

Practically, the research may provide simplified candidate parameterisations or diagnostic relationships that can support land surface model development. Even where a discovered equation is not suitable as a direct replacement for a process-based scheme, the selected terms and their stability may reveal useful hypotheses about dominant controls on observed fluxes. The project will also produce a reproducible computational workflow using Python, high-performance computing and GitHub, thereby supporting transparent and collaborative research practice.

1.8 Scope and Delimitation of the Study

The study will focus on equation discovery for selected land–atmosphere fluxes and their relationship with meteorological forcings and relevant environmental variables available through FLUXNET products. Depending on data completeness and the final analytical design, target variables may include latent heat flux, sensible heat flux, net ecosystem exchange or other closely related flux products. The study will not attempt to discover a complete land surface model. Instead, it will examine selected functional relationships and evaluate whether sparse symbolic equations can represent them adequately.

The candidate library will be physically informed and may include linear, nonlinear, interaction, threshold and saturation-like terms derived from variables such as incoming radiation, temperature, vapour pressure deficit, soil water status, precipitation and vegetation-related indicators. The project will compare sites and groups of sites where data availability and quality are sufficient. Because observational coverage is uneven and ecosystem processes differ across regions, conclusions will be restricted to the sampled data and should not be interpreted as universal governing laws.

1.9 Operational Definition of Key Terms

Equation discovery: a data-driven process for identifying explicit mathematical equations that describe observed relationships or system dynamics.

SINDy: Sparse Identification of Nonlinear Dynamics, a framework that discovers a small number of active terms from a larger library of candidate nonlinear functions using sparse regression.

Land–atmosphere flux: the exchange of energy, water or carbon between the land surface and the atmosphere.

Parsimonious model: a model that explains data adequately while using relatively few terms or parameters.

Interpretability: the extent to which a model's structure and behaviour can be understood, inspected and related to scientific knowledge.

Generalisation: the ability of a model or discovered equation to maintain useful performance on data not used during model development.

CHAPTER TWO

LITERATURE REVIEW

2.1 Conceptual Overview of Land–Atmosphere Exchange

Land–atmosphere coupling links terrestrial ecosystems to the climate system through the exchange of radiation, heat, moisture and carbon. At the surface, available energy is partitioned among sensible heat, latent heat and ground heat storage, while vegetation and soils regulate water and carbon exchanges. These processes depend on both atmospheric demand and land-surface supply. For example, latent heat exchange is strongly influenced by available energy, vapour pressure deficit, aerodynamic conditions and water availability, while carbon uptake is affected by radiation, temperature, atmospheric conditions, phenology and plant water stress.

Because these controls interact, simple single-variable relationships are often insufficient. A response to temperature may depend on soil moisture; a radiation response may saturate; and atmospheric dryness may exert different effects under wet and dry conditions. These nonlinearities provide a rationale for machine-learning approaches but also require careful model design. An equation-discovery method is particularly attractive when it can represent interactions and nonlinear terms without sacrificing the ability to inspect the final functional form.

2.2 Machine Learning in Earth System and Geoscience Modelling

Machine learning has become increasingly important in Earth system science because large observational and simulation datasets create opportunities for data-driven model development. However, purely data-driven methods may struggle with extrapolation and

physical consistency, especially when deployed under conditions outside their training domain. Reviews of physics-informed and differentiable approaches emphasise the value of combining scientific constraints with flexible learning systems (Shen et al., 2023). The central idea is not that machine learning should replace physical understanding, but that prior knowledge can constrain the learning problem and reduce the space of implausible solutions.

For the present study, physical knowledge will enter primarily through candidate-library design and model evaluation. Candidate terms will be selected because they correspond to plausible controls or interactions rather than simply because they improve in-sample fit. Physical plausibility will also be assessed after equation discovery. This distinction is important because an equation can fit observed data while still being scientifically unreasonable or unstable under extrapolation.

2.3 Equation Discovery and Sparse Modelling

Equation discovery aims to infer mathematical structure from data. Earlier symbolic-regression approaches searched broadly over possible mathematical expressions, whereas modern sparse-regression methods exploit the observation that many physical systems can be represented by a relatively small number of active terms drawn from a larger candidate library. Brunton et al. (2016) formalised this idea in SINDy by combining nonlinear feature libraries with sparse regression.

In a generic SINDy formulation, observations of a system are arranged into a data matrix X and a library of candidate functions $\Theta(X)$. Sparse regression is then used to estimate a coefficient vector Ξ such that the observed system behaviour is represented by a small number of active candidate terms. Conceptually, the problem can be written as $\dot{X} \approx \Theta(X)\Xi$ for dynamical systems, or adapted to other response formulations where the objective

is to identify a sparse functional relationship. The key modelling decision is therefore not only the regression algorithm but also the construction of the candidate library.

Parsimony is a central advantage of this framework. A sparse model is easier to inspect and can reduce overfitting, but sparsity also introduces risk: an overly restrictive penalty may exclude a real process, while an insufficient penalty may retain many unstable terms. Therefore, threshold selection and model stability must be evaluated systematically rather than chosen only by visual inspection.

2.4 The SINDy Approach

Standard SINDy begins with measurements of relevant variables and, where required, their time derivatives. Candidate functions may include constants, linear terms, polynomial terms, interactions, trigonometric functions or other basis functions. Sparse regression then seeks a coefficient matrix in which most entries are zero. Sequentially thresholded least squares and related sparse optimisers are commonly used because they balance computational efficiency with model sparsity.

The appeal of SINDy for land–atmosphere modelling is that the candidate library can encode scientific expectations. Rather than allowing an unrestricted algorithm to search through arbitrary functions, the library can include terms motivated by surface-energy and ecosystem processes. Examples may include radiation terms, temperature-response terms, moisture constraints, interactions between atmospheric demand and water availability, and nonlinear saturation functions. The final equation can therefore be interpreted in relation to known physical mechanisms.

Nevertheless, SINDy was originally developed primarily for discovering dynamical equations from relatively clean measurements. Environmental observations create additional

difficulties. Measurements contain random and systematic errors; missing values may be common; variables can be highly correlated; and relationships can change across seasons, ecosystems and spatial scales. These characteristics make direct transfer of standard SINDy methods uncertain and justify careful methodological testing.

2.5 FLUXNET Observations and Their Relevance

FLUXNET integrates eddy-covariance observations from regional networks to provide a major source of ecosystem-scale measurements of carbon, water and energy exchange. The FLUXNET2015 dataset and the ONEFlux processing pipeline provide harmonised products with extensive processing, quality-control and uncertainty information (Pastorello et al., 2020). The dataset contains a wide range of meteorological and flux variables that are suitable for investigating empirical process relationships.

The strengths of FLUXNET include temporal richness, ecosystem diversity and direct relevance to land–atmosphere exchange. However, the data also contain limitations that are important for equation discovery. Tower footprints are site-specific; measurement gaps occur; quality-control procedures affect data availability; and observed relationships may reflect both local processes and measurement conditions. Furthermore, different sites represent different climates, vegetation structures and management histories. A discovered equation must therefore be evaluated across independent sites rather than judged solely by performance on randomly mixed observations.

2.6 Interpretability, Physical Plausibility and Generalisation

Interpretability is not guaranteed merely because a model produces a symbolic equation. An equation may be compact but still difficult to justify physically. The proposed

study therefore distinguishes structural interpretability from scientific plausibility. Structural interpretability concerns whether the equation is short enough to inspect. Scientific plausibility concerns whether its terms, signs, response shapes and limiting behaviour are consistent with physical understanding.

Generalisation is equally important. A model that performs well on observations from the same site and season may fail when applied to a different biome or climate regime. Site-wise and biome-wise validation are therefore more scientifically meaningful than random train-test splitting alone. The study will use multiple validation strategies to distinguish interpolation within a sampled distribution from transfer to genuinely different environmental conditions.

2.7 Empirical Review

Brunton et al. (2016) demonstrated that sparse regression can recover compact governing equations from data across a range of nonlinear systems. Their work established SINDy as a practical framework for balancing candidate-library richness with sparse model selection. Subsequent work has extended the approach to coordinate discovery, partial differential equations, control and noisy data settings (Champion et al., 2019; Messenger et al., 2022).

Champion et al. (2019) showed that data-driven discovery can involve not only identification of equations but also the discovery of useful coordinates in which governing relationships become sparse. This is relevant to environmental systems because the observed variables may not always provide the most compact representation of the underlying dynamics. However, coordinate discovery can reduce direct interpretability if transformed variables lose an obvious physical meaning. The present study will therefore prioritise

physically meaningful observed variables and derived quantities before considering more abstract transformations.

Messenger et al. (2022) demonstrated that ensemble methods can improve the robustness of sparse model discovery under limited and noisy data conditions. Their work is especially relevant to environmental observations because it quantifies the stability with which candidate terms are selected across resampled datasets. Rather than relying on one fitted sparse model, ensemble approaches can estimate inclusion probabilities and uncertainty. This offers a useful strategy for determining whether a term is consistently supported by the available observations.

Recent atmospheric research has also shown the potential of weak-form SINDy to derive physically interpretable atmospheric models from simulated and assimilated data (Minor et al., 2025). This development demonstrates that sparse equation discovery can be adapted to high-dimensional geophysical data, while also reinforcing the importance of robust formulations for noisy spatiotemporal systems. The present study extends this broader line of inquiry to land–atmosphere exchange using site-level observational data.

Overall, the literature supports the scientific promise of equation discovery but also highlights an unresolved gap. The success of the method depends heavily on data quality, feature-library design, optimisation choices and validation strategy. There is therefore a need for a focused study that evaluates these issues explicitly within land surface modelling rather than treating equation discovery as a generic algorithmic exercise.

2.8 Theoretical Framework

The study is guided by three complementary ideas: sparse modelling, physical consistency and empirical validation. First, sparse modelling assumes that a useful

description may be represented by a relatively small subset of candidate terms. Second, physical consistency requires that candidate terms and discovered relationships should be evaluated against established scientific expectations. Third, empirical validation requires that equations be tested on observations not used during fitting, particularly across sites and environmental regimes. Together, these principles form a framework in which predictive accuracy alone is not the criterion for success.

The conceptual relationship is as follows: meteorological and environmental observations provide the empirical basis; scientific understanding guides construction of the candidate library; SINDy performs sparse selection; the discovered equation is evaluated for predictive skill, sparsity, stability and physical plausibility; and cross-site validation determines whether the equation generalises. Where the method fails, the failure itself is treated as evidence concerning the limits of equation discovery in this application domain.

2.9 Gap in the Literature

Existing studies establish SINDy as a powerful method for discovering equations from data and demonstrate growing applications in atmospheric and geophysical science. However, the application of sparse equation discovery specifically to observational land–atmosphere flux relationships remains underexplored. There is insufficient evidence on how candidate-library choices, observational noise, data gaps and environmental heterogeneity affect the stability and scientific usefulness of discovered equations. The present study addresses this gap through a structured, reproducible and comparative analysis using FLUXNET observations.

CHAPTER THREE

METHODOLOGY

3.1 Research Design

The study will adopt a quantitative, computational and comparative research design. It will use secondary observational data from the FLUXNET data infrastructure and apply machine learning-based equation discovery to selected land–atmosphere fluxes. The design is explanatory and methodological: it seeks both to identify equations and to evaluate the conditions under which the discovery process produces stable and scientifically useful results.

The analytical workflow will consist of six main stages: data acquisition and screening; preprocessing and quality control; physically informed feature construction; equation discovery; model validation and comparison; and sensitivity and robustness analysis. The study will be implemented as a reproducible computational pipeline using Python. Computationally demanding experiments, including repeated cross-validation and ensemble analyses, will be suitable for execution on high-performance computing resources.

3.2 Data Source

The primary data source will be the FLUXNET observational data products, subject to the applicable data policy and access conditions. The FLUXNET2015 dataset and associated ONEFlux processing system provide harmonised meteorological and eddy-covariance data products from sites representing diverse ecosystems and climatic conditions (Pastorello et al., 2020). Site selection will be based on data completeness, temporal

coverage, availability of required variables and representation across contrasting environmental conditions.

Relevant predictor variables are expected to include combinations of incoming shortwave radiation, air temperature, vapour pressure deficit, precipitation, wind-related variables and other meteorological or environmental indicators available in the selected product. Target variables will be chosen according to data quality and scientific relevance and may include latent heat flux, sensible heat flux and net ecosystem exchange or related flux products. Final variable selection will be documented transparently before model fitting.

3.3 Site Selection Strategy

A stratified site-selection strategy will be used where possible. Sites will first be screened for sufficient record length and data completeness. The remaining sites will then be grouped by broad vegetation or climate categories to ensure that the analysis does not rely on a single ecosystem type. Separate experiments may be conducted for pooled datasets and for site-specific or biome-specific subsets.

The study will avoid relying exclusively on random observation-level splits because observations from the same site are temporally autocorrelated and may share local characteristics. Instead, validation will include blocked temporal splits and leave-site-out or group-wise validation. This approach will provide a stronger test of whether discovered equations can transfer beyond the site conditions represented in model training.

3.4 Data Preprocessing

Data preprocessing will include inspection of quality flags, removal or exclusion of observations that do not meet predefined quality criteria, treatment of missing values, unit

harmonisation and construction of consistent temporal resolutions. The exact treatment of gaps will depend on the experiment. Baseline analyses will use complete observations after quality screening, while sensitivity experiments may compare alternative gap-handling procedures.

Variables will be inspected for outliers, physically impossible values and distributional extremes. Transformations will be applied only where they have a clear methodological justification. Scaling may be required during regression, particularly when candidate terms have substantially different magnitudes. However, coefficients and equations will be transformed back or reported in a form that allows physical interpretation where possible.

Temporal aggregation will be tested as a methodological factor. Half-hourly observations preserve high-frequency information but may contain stronger measurement noise, while daily or other aggregated values can improve signal-to-noise ratios at the cost of temporal detail. The study will evaluate whether the stability of equation discovery changes across selected temporal resolutions.

3.5 Candidate Feature Library

The candidate library will be constructed from both mathematical flexibility and physical reasoning. Candidate terms may include: (a) first-order meteorological controls; (b) polynomial responses where nonlinear curvature is plausible; (c) interaction terms representing coupled controls; (d) saturation or bounded-response terms where justified; and (e) transformed variables with clear scientific meaning. The final library will be deliberately documented because feature-library specification is a major determinant of equation-discovery outcomes.

An illustrative library for a flux target Y may contain terms based on radiation R , temperature T , vapour pressure deficit D and a water-availability variable W , such as $1, R, T, D, W, R^2, T^2, R \times D, R \times W, D \times W$ and selected nonlinear response functions. This does not imply that all terms are physically correct. Rather, they represent candidate relationships from which sparse regression can select a smaller combination.

To assess the value of scientific prior knowledge, at least two library strategies will be compared where feasible: a physically informed library and a broader generic nonlinear library. The comparison will determine whether physical feature design improves sparsity, stability or transferability.

3.6 Equation Discovery Procedure

The main equation-discovery framework will be SINDy. The PySINDy software ecosystem provides an open-source implementation of SINDy and related model-discovery methods and will be considered as a core implementation platform. Sparse optimisers and threshold parameters will be tuned using training and validation procedures designed to avoid information leakage.

For each target flux, candidate equations will be generated across a grid or structured search of library specifications, sparsity thresholds and optimisation settings. Models will be retained and compared using a multi-criteria framework that considers predictive error, number of active terms, coefficient stability and physical plausibility. A model will not be selected solely because it has the lowest in-sample error.

Where standard SINDy requires derivatives and derivative estimation is unstable, alternative formulations will be explored. Depending on the structure of the target problem, these may include smoothing before differentiation, integral or weak-form approaches, or

formulations that directly discover sparse input-output relationships. The purpose is to distinguish limitations of the scientific problem from limitations introduced by a particular numerical implementation.

3.7 Baseline Models

To determine whether equation discovery provides practical value, SINDy models will be compared with baseline approaches. Baselines may include multiple linear regression, regularised regression and selected nonlinear machine-learning models such as random forests or gradient-boosted trees. The exact set will be limited to models that provide meaningful comparison rather than maximising the number of algorithms.

The comparison will focus on the trade-off between accuracy and interpretability. A black-box model may achieve lower prediction error but offer limited mechanistic insight. Conversely, a sparse equation may be scientifically attractive even if its predictive performance is slightly weaker. Therefore, the analysis will report both predictive and structural criteria rather than declaring one model universally superior.

3.8 Model Evaluation

Predictive performance will be assessed using metrics appropriate to continuous targets, including root mean square error (RMSE), mean absolute error (MAE), coefficient of determination (R^2) and bias-related diagnostics where appropriate. Metrics will be reported separately for training, validation and independent test partitions.

Equation complexity will be evaluated using the number of active terms, coefficient magnitude and model description length where suitable. Stability will be evaluated by repeated resampling or repeated fitting under small changes in training data and

hyperparameters. A stable term is one that is repeatedly selected with similar sign and magnitude under comparable modelling conditions.

Physical plausibility will be assessed using expert-informed criteria. These may include expected coefficient signs, boundedness or sensible behaviour within the observed range, response curves under controlled changes in individual drivers, dimensional consistency where possible and consistency with known limiting conditions. Such checks are important because sparse regression does not automatically guarantee physical correctness.

3.9 Robustness and Sensitivity Analysis

Robustness experiments will systematically test the effect of methodological choices. Planned factors include candidate-library size, sparsity threshold, temporal resolution, measurement-noise level, missing-data treatment and site composition. Synthetic perturbations may be added to selected experiments to quantify how quickly model structure changes as noise increases.

An ensemble strategy will be considered to estimate term-selection frequencies across bootstrap or subsampled datasets. Following the principles demonstrated by Messenger et al. (2022), inclusion frequency can provide a useful measure of confidence in the structural components of a discovered equation. Terms that appear only under a narrow set of training samples or hyperparameters will be treated as less reliable.

Generalisation will be evaluated through site-wise and group-wise testing. A particularly important experiment will involve training on selected sites and testing on held-out sites or biomes. This will reveal whether an equation captures broader relationships or merely fits site-specific correlations.

3.10 High-Performance Computing Strategy

High-performance computing will be used for experiments that require repeated model fitting across many sites, candidate libraries and hyperparameter combinations. Independent experiments will be parallelised where possible, including site-level analyses, cross-validation folds, bootstrap replicates and sensitivity scenarios. Parallel execution will reduce wall-clock time and make broader robustness analysis feasible.

The workflow will record software versions, configuration files, random seeds and job parameters. Resource usage, failed jobs and runtime behaviour will also be monitored so that computational limitations can be reported as part of the research findings rather than ignored.

3.11 Software Engineering and Reproducibility

The project will be implemented primarily in Python. Core tasks will include data processing, numerical analysis, sparse regression, visualisation and experiment management. GitHub will be used for version control, issue tracking and collaborative development. The repository will separate raw-data access instructions from processed outputs where data-sharing restrictions apply.

A reproducible project structure will include directories for source code, configuration files, tests, documentation, results and figures. Automated checks will be used where feasible. Every major experiment will be driven by recorded configuration parameters so that results can be reproduced without manually editing analysis code. This software structure is especially important because equation-discovery outcomes may be sensitive to seemingly minor methodological changes.

3.12 Ethical and Data Governance Considerations

The study will use environmental observational data rather than human participants and is therefore expected to involve minimal direct human-subject risk. Nevertheless, data access, attribution and acknowledgement requirements will be followed carefully. FLUXNET data use will comply with the applicable data policy, including any requirements concerning site acknowledgement and redistribution of data products.

Reproducibility will be balanced with data-governance restrictions. Where redistribution of raw data is not permitted, the public repository will provide code, metadata, processing instructions and configuration files necessary for authorised users to reproduce the workflow after obtaining data through the appropriate channel.

3.13 Expected Limitations

Several limitations are anticipated. First, SINDy may be sensitive to measurement noise and to the numerical estimation of derivatives. Second, candidate-library design introduces researcher judgement and can bias the discovered structure. Third, observational data may not contain sufficient information to distinguish causation from correlation. Fourth, land–atmosphere relationships can be regime-dependent, meaning that no single equation may generalise across all ecosystems. Fifth, a sparse equation may represent an effective approximation of observed behaviour rather than a literal governing law.

These limitations will not be treated merely as obstacles. They are central to the research objective because the project explicitly aims to identify the circumstances under which equation discovery succeeds and fails.

3.14 Expected Outcomes

A curated and reproducible workflow for preparing selected FLUXNET variables for equation discovery.

A physically informed candidate-library framework for land–atmosphere flux relationships.

A set of sparse candidate equations for selected flux targets, accompanied by uncertainty and stability diagnostics.

A comparative evaluation of equation discovery and selected baseline predictive models.

Evidence on the effects of noise, missing data, library design and environmental heterogeneity.

Recommendations on when SINDy-style equation discovery is scientifically useful for land surface modelling and when more robust or alternative approaches are required.

A version-controlled Python research repository designed for reproducibility and collaborative development.

3.15 Proposed Work Plan

Phase	Main Activities	Indicative Duration
1	Literature review, research design and data familiarisation	Months 1–2
2	Data access, site screening and preprocessing pipeline	Months 2–3
3	Candidate-library design and baseline modelling	Months 3–4
4	SINDy implementation and initial equation discovery	Months 4–6
5	Robustness, ensemble and cross-site experiments	Months 6–8
6	Interpretation, validation and computational scaling	Months 8–9
7	Writing, revision, documentation and final repository	Months 9–10

3.16 Summary of Methodological Logic

The methodology is designed to test equation discovery as a scientific method rather than merely as a machine-learning benchmark. FLUXNET observations will provide empirical evidence; physical knowledge will constrain the candidate terms; SINDy will perform sparse equation selection; baseline models will provide predictive context; and repeated site-wise and sensitivity experiments will determine whether the resulting equations are stable and transferable. The final interpretation will therefore consider four questions simultaneously: Does the equation predict? Is it sparse? Is it stable? Is it scientifically plausible?

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